

# CV Maker — Architecture Analysis

---

**Project:** CV Maker

**Technology:** Python/FastAPI backend, React/Vite + Svelte frontends, PostgreSQL, SQLAlchemy, Playwright scraper

**Location:** /home/dominic/Documents/CV MAKER

**Report #:** 2/20

## Architecture Type

---

Full-stack monorepo with separate backend and two frontends

## Layers

---

Frontend (React/Svelte) → REST API → Service Layer → SQLAlchemy → PostgreSQL + Playwright Scraper

## Design Patterns

---

Service layer pattern, repository pattern for DB access

## Scalability

---

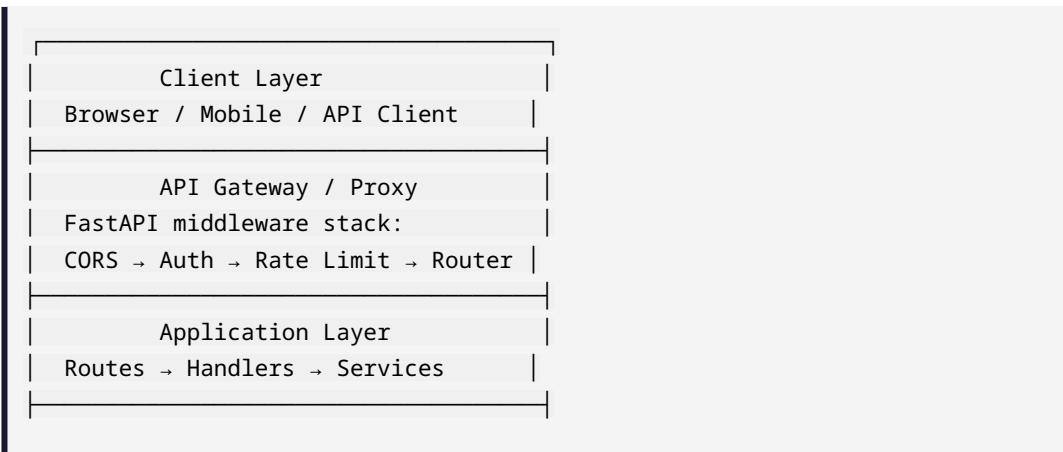
Can scale horizontally with PostgreSQL backend; frontends are decoupled

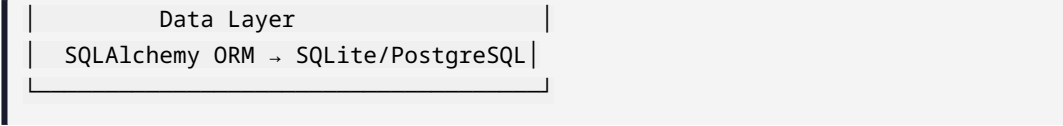
---

## System Architecture

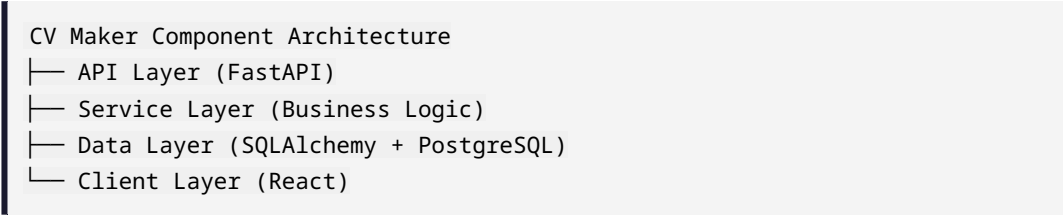
---

### Layer Architecture





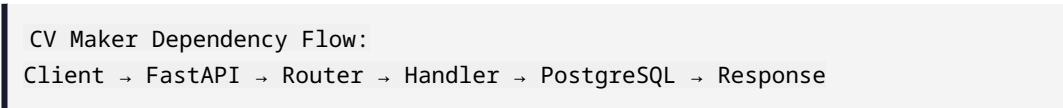
Component Interaction



Request Lifecycle

- 1. **HTTP Request** arrives at FastAPI
- 2. **CORS middleware** checks origin headers
- 3. **Auth dependency** extracts and validates JWT
- 4. **Rate limiter** checks request count per IP
- 5. **Router** matches path to handler
- 6. **Handler** executes business logic
- 7. **Database session** commits or rolls back
- 8. **Response** returned to client

Dependency Graph



Design Patterns Used

| Pattern              | Location           | Purpose                |
|----------------------|--------------------|------------------------|
| Dependency Injection | FastAPI Depends()  | DB sessions, auth      |
| Repository Pattern   | SQLAlchemy models  | Data abstraction       |
| Middleware Chain     | FastAPI middleware | Cross-cutting concerns |
| Router Module        | APIRouter include  | Separation of concerns |